

ELEC5618 Assignment 3 — Unit Test List

Shattered Pixel Dungeon v3.0.2 | 75 Tests Total

Improvement 1 — In-Game Screenshot Shortcut

13 tests

ScreenshotFeatureTest.java

- 1 Screenshot class exists in the codebase
- 2 capture() method is public and static
- 3 Filename matches the expected format (screenshot_YYYY-MM-DD_HH-mm-ss.png)
- 4 Filename starts with 'screenshot_'
- 5 Filename ends with '.png'
- 6 Timestamp in filename is a valid date
- 7 Multiple captures produce unique filenames
- 8 Screenshot file is readable after being written
- 9 Written file name matches the naming pattern
- 10 Background thread can be created and started
- 11 Capture does not block the main game thread
- 12 Vertical flip reverses pixel row order correctly
- 13 Vertical flip middle rows are swapped correctly

Improvement 2 — Actor/Damage System Refactoring

45 tests

DamagePropertyTest.java

- 1 DamageProperty enum contains all expected values
- 2 All enum value names are valid and non-empty
- 3 EnumSet works correctly with DamageProperty values
- 4 Empty EnumSet (noneOf) behaves correctly
- 5 Char class has both 2-parameter and 3-parameter damage() methods
- 6 DamageProperty is in the correct package
- 7 New properties can be added without breaking existing code

CharDamagePropertyTest.java

- 1 3-parameter damage() method exists with correct signature
- 2 2-parameter damage() method exists for backwards compatibility
- 3 Both damage() methods are correctly overloaded
- 4 DamageProperty enum is structured correctly
- 5 IGNORES_SHIELDS value exists in the enum
- 6 DamageProperty works correctly with EnumSet

7	Code is maintainable (public enum, correct package)
8	New properties can be combined dynamically
9	Old 2-parameter method delegates to new 3-parameter method
10	Refactoring does not remove the old damage() method
11	Both damage() methods belong to the Char class
12	Edge cases handled (null sets, empty sets, full sets)

DamageCalculatorTest.java

1	Positive damage with no properties returns unchanged
2	Zero damage returns zero
3	Negative damage returns zero
4	TRUE_DAMAGE returns base damage unmodified
5	IGNORES_SHIELDS does not alter the damage number
6	IGNORES_RESISTANCE does not alter the damage number
7	Null property set does not crash
8	Empty set does not bypass shields
9	IGNORES_SHIELDS bypasses shields
10	TRUE_DAMAGE bypasses shields
11	UNBLOCKABLE alone does not bypass shields
12	IGNORES_RESISTANCE alone does not bypass shields
13	Null set does not bypass shields
14	Empty set does not bypass resistance
15	IGNORES_RESISTANCE bypasses resistance
16	TRUE_DAMAGE bypasses resistance
17	IGNORES_SHIELDS alone does not bypass resistance
18	Null set does not bypass resistance
19	Empty set is not unblockable
20	UNBLOCKABLE property is unblockable
21	TRUE_DAMAGE is unblockable
22	IGNORES_SHIELDS alone is not unblockable
23	Null set is not unblockable
24	TRUE_DAMAGE_SET bypasses shields, resistance, and block all at once
25	IGNORES_SHIELDS_SET only bypasses shields (not resistance or block)
26	DamageCalculator cannot be instantiated (utility class)

Improvement 3 — Crash-Safe Save Backup and Recovery

17 tests

SaveBackupRecoveryTest.java

1	Backup filename has .bak suffix
2	Depth file backup follows same naming convention
3	Primary save file is detected when present
4	Primary save file is detected as missing when absent
5	Backup file is created alongside the primary file
6	Backup file content matches the primary file
7	Backup file is overwritten on subsequent saves
8	Corrupted primary file triggers backup recovery
9	Primary file is restored from backup after corruption
10	Missing primary file triggers backup recovery
11	Primary file is created from backup when missing
12	Recovery fails cleanly when both files are missing
13	Recovery fails cleanly when backup is also corrupt
14	Recovery fails cleanly when backup is empty
15	Recovering slot 1 does not affect slot 2
16	Depth save files (depth1.dat) can also be recovered
17	Recovery log messages use the correct [SAVE RECOVERY] prefix
Improvement 1	13 tests
Improvement 2	45 tests
Improvement 3	17 tests
Total	75 tests